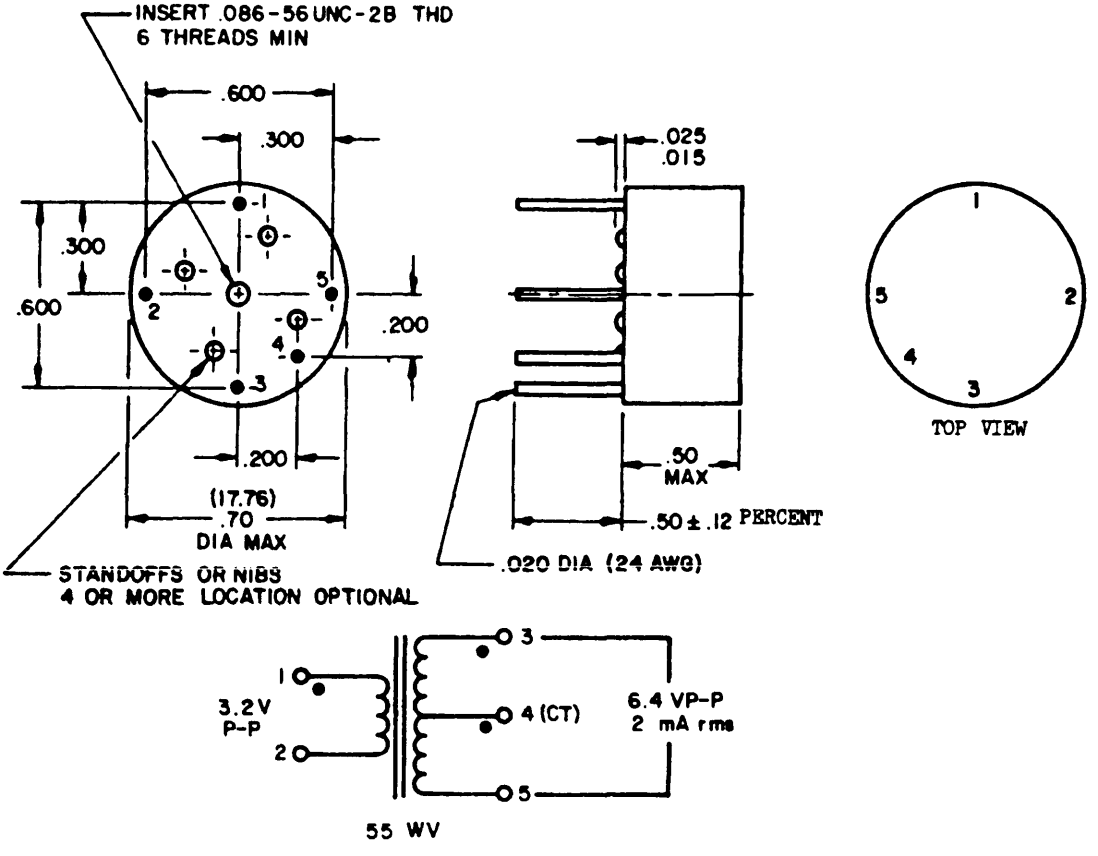
MILITARY SPECIFICATION SHEET

TRANSFORMER, POWER, STEP-UP, TF5V03ZZ317

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the transformers described herein shall consist of this document and the latest issue of Specification MIL-T-27.



CIRCUIT DIAGRAM AND MARKING

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Metric equivalents are in parentheses.
4. Tolerance on terminal position dimensions is ±.010.
5. Marking and terminal location shall be on top of the case.

FIGURE 1. Dimensions and configuration.

FSC 5950

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REQUIREMENTS: (When numbers in parentheses, i.e., (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings:

Primary voltage,
(1-2): 3.2 volts P-P, 12.5 kHz, square wave terminal.

Secondary voltage,
(3-5): 6.4 volts, P-P.

Working voltage,
(1-2): 55 volts maximum, 25 kHz.
(3-5): 55 volts maximum, 25 kHz.

Design and construction:

Dimensions and configuration: See figure 1.

Duty cycle: Continuous.

Case: Encapsulated.

Material: Epoxy.

Terminals: Pin, N-2 type, gold plated nickel, in accordance with MIL-STD-1276.
Terminal height: .5 inch ±.12 percent.

Weight: 8 grams maximum.

Operating temperature range: -55°C to 155°C.

Terminal strength: Method 211, MIL-STD-202, test condition A, 2 pounds.

Dielectric withstanding voltage (at sea level): 300 volts rms.

Electrical characterisitcs:

(1-2) Primary voltage for test purpose: 12.5 kHz ±5 percent sine wave voltage of 1.13 volts rms ±2 percent.

Primary exitation current: 4 milliamperes rms maximum.

Maximum direct current: .25 milliampere.

Secondary current: 2 milliamperes maximum.

Secondary voltage: 2.26 volts ±2 percent, 2 milliamperes rms full load.

Secondary centertap accuracy: ±1 percent.

Secondary dc unbalance: .1 milliampere maximum.

DC resistance measured at 25°C:
Primary (1-2): 1.45 ohms maximum.
Secondary (3-5): 3.40 ohms maximum.

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Temperature rise: 10°C maximum.

Salt spray: Test in accordance with Method 101, MIL-STD-202.

Polarity: Additive with terminals 2 and 3 connected.

Vibration: Method 204, MIL-STD-202, test condition A.

Shock: Method 213, MIL-STD-202, test condition I, 6 milliseconds duration.

Marking location: See figure 1.

Part number: M27/31-01.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - EL, MI, SF
Navy - EC, SH, OS
Air Force - 11, 17, 80
DSA - ES

User activities:
Army - MU, WC, SG, ME
Navy - MC, AS
Air Force - 19

Preparing activity:
Army - EL

Agent:
DSA - ES

(Project 5950-0500-1)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-T-27/31 TRANSFORMER, POWER, STEP-UP, TF5V03ZZ317

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-PM 1068-68

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Electric Boat Division